

#1. To write a Python program to add two numbers entered by the user.

```
a = float(input("Enter first number: "))
b = float(input("Enter second number: "))

sum = a + b

print("Sum =", sum)
```

```
Enter first number: 7
Enter second number: 9
Sum = 16.0
```

#2. To write a Python program to interchange the values of two variables.

```
a = input("Enter first value: ")
b = input("Enter second value: ")

a, b = b, a

print("After swapping:")
print("a =", a)
print("b =", b)
```

```
Enter first value: 6
Enter second value: 7
After swapping:
a = 7
b = 6
```

#3. To write a Python program to find the largest among three numbers using conditional statements.

```
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))
c = int(input("Enter third number: "))

if a >= b and a >= c:
    print("Largest =", a)
elif b >= a and b >= c:
    print("Largest =", b)
else:
    print("Largest =", c)
```

```
Enter first number: 5
Enter second number: 7
Enter third number: 31
Largest = 31
```

#4. To write a Python program to check whether a given number is even or odd.

```
num = int(input("Enter a number: "))

if num % 2 == 0:
    print("Even")
else:
    print("Odd")
```

```
Enter a number: 16
Even
```

#5. To write a Python program to determine whether a number is positive, negative, or zero.

```
num = int(input("Enter a number: "))

if num > 0:
    print("Positive")
elif num < 0:
    print("Negative")
else:
    print("Zero")
```

```
Enter a number: 21
Positive
```

#6. To write a Python program to check whether a given year is a leap year.

```
year = int(input("Enter year: "))

if (year % 400 == 0) or (year % 4 == 0 and year % 100 != 0):
    print("Leap Year")
else:
    print("Not a Leap Year")
```

Enter year: 2004
Leap Year

#7. To write a Python program to perform basic arithmetic operations using operators.

```
a = float(input("Enter first number: "))
b = float(input("Enter second number: "))

print("Addition =", a + b)
print("Subtraction =", a - b)
print("Multiplication =", a * b)
print("Division =", a / b)
```

Enter first number: 31
Enter second number: 34
Addition = 65.0
Subtraction = -3.0
Multiplication = 1054.0
Division = 0.9117647058823529

#8. To write a Python program to display the multiplication table of a given number.

```
num = int(input("Enter a number: "))

for i in range(1, 11):
    print(num, "x", i, "=", num * i)
```

Enter a number: 9
9 x 1 = 9
9 x 2 = 18
9 x 3 = 27
9 x 4 = 36
9 x 5 = 45
9 x 6 = 54
9 x 7 = 63
9 x 8 = 72
9 x 9 = 81
9 x 10 = 90

#9. To write a Python program to calculate the factorial of a number using a loop.

```
num = int(input("Enter a number: "))

fact = 1

for i in range(1, num + 1):
    fact = fact * i

print("Factorial =", fact)
```

Enter a number: 5
Factorial = 120

#10. To write a Python program to find the sum of the first N natural numbers.

```
n = int(input("Enter N: "))

sum = 0

for i in range(1, n + 1):
    sum += i

print("Sum =", sum)
```

Enter N: 6
Sum = 21

#11. To write a Python program to print numbers from 1 to N using a for loop.

```
n = int(input("Enter N: "))

for i in range(1, n + 1):
    print(i)
```

Enter N: 14

1
2
3
4
5
6
7
8
9
10
11
12
13
14

#12. To write a Python program to print numbers in reverse order from N to 1.

```
n = int(input("Enter N: "))

for i in range(n, 0, -1):
    print(i)
```

Enter N: 10

10
9
8
7
6
5
4
3
2
1

#13. To write a Python program to print all even numbers up to a given limit.

```
n = int(input("Enter limit: "))

for i in range(2, n + 1, 2):
    print(i)
```

Enter limit: 12

2
4
6
8
10
12

#14. To write a Python program to reverse the digits of a given number.

```
num = int(input("Enter a number: "))

rev = 0

while num > 0:
    digit = num % 10
    rev = rev * 10 + digit
    num = num // 10

print("Reversed Number =", rev)
```

Enter a number: 657
Reversed Number = 756

#15. To write a Python program to count the number of digits in a given integer.

```
num = int(input("Enter a number: "))
```

```
count = 0

while num > 0:
    count += 1
    num = num // 10

print("Number of digits =", count)
```

Enter a number: 345678
Number of digits = 6

#16. To write a Python program to calculate the sum of digits of a given number.

```
num = int(input("Enter a number: "))

sum = 0

while num > 0:
    digit = num % 10
    sum += digit
    num = num // 10

print("Sum of digits =", sum)
```

Enter a number: 432
Sum of digits = 9

#17. To write a Python program to check whether a given number is prime.

```
num = int(input("Enter a number: "))

if num > 1:
    for i in range(2, num):
        if num % i == 0:
            print("Not Prime")
            break
    else:
        print("Prime")
else:
    print("Not Prime")
```

Enter a number: 477
Not Prime

#18. To write a Python program to generate the Fibonacci series up to N terms.

```
n = int(input("Enter number of terms: "))

a = 0
b = 1

for i in range(n):
    print(a)
    c = a + b
    a = b
    b = c
```

Enter number of terms: 13
0
1
1
2
3
5
8
13
21
34
55
89
144

#19. To write a Python program to check whether a string is a palindrome.

```
text = input("Enter a string: ")

if text == text[::-1]:
```

```
    print("Palindrome")
else:
    print("Not Palindrome")
```

Enter a string: radar
Palindrome

20. To write a Python program to check whether a given number is an Armstrong number.

```
num = int(input("Enter a number: "))

temp = num
digits = len(str(num))
sum = 0

while temp > 0:
    digit = temp % 10
    sum += digit ** digits
    temp = temp // 10

if sum == num:
    print("Armstrong Number")
else:
    print("Not an Armstrong Number")
```

Enter a number: 34
Not an Armstrong Number